EE3-01 Analogue integrated circuits + systems

Answers to question 1:

1.a i) Bookwork

Student must explain they are self-biased current references and explain the need and role of the startup circuit based on equations. [4 points]

Equation for output current:

$$I_O = \frac{(V_{be1} - V_{be2})}{R} = \frac{V_T \ln\left[\frac{I_{IN}}{I_O}\right]}{R} = \frac{V_T}{R}\ln\left[\frac{I_{IN}}{I_{S1}} \times \frac{I_{S2}}{I_O}\right]$$

If we self bias and force $I_{IN}/I_O = 1$ with top current mirror, then ln(1) = 0

Require $I_{IN}/I_O > 1$

In Figure 1.1, $I_{IN}/I_O = n = 2$

So, assuming matched devices

$$I_O = \frac{V_T}{R}\ln[2]$$

[Total 4 points]

Must elaborate that the circuit in Fig. 1.1 has a better temperature coefficient that that of fig 1.2 and show equations for Temperature coefficient.

Fig 1.1 is a Improved TCF using VT Referenced Current Source/Sink

$$TC_F = \left(\frac{1}{V_T}\right)\frac{\delta V_T}{\delta T} - \left(\frac{1}{R}\right)\frac{\delta R}{\delta T}$$ [1 point]

Fig 1.2 is a V_{be} Referenced Sink/Source with a temperature coefficient:

$$TC_F = \left(\frac{1}{V_{be1}}\right)\frac{\delta V_{be1}}{\delta T} - \left(\frac{1}{R}\right)\frac{\delta R}{\delta T}$$ [1 point]

[Total 6 points]

b) For Fig 1.1:

$$TC_F = \frac{1}{T} - \left(\frac{1}{R}\right)\frac{\delta R}{\delta T}$$

$$= \left(\frac{1}{300} - 1500x10^{-6}\right) = 1833 ppm/^\circ C$$

[3 points]

For Fig 1.2

$$TC_F = \left(\frac{1}{V_{be1}}\right)\frac{\delta V_{be1}}{\delta T} - \left(\frac{1}{R}\right)\frac{\delta R}{\delta T}$$

Solutions

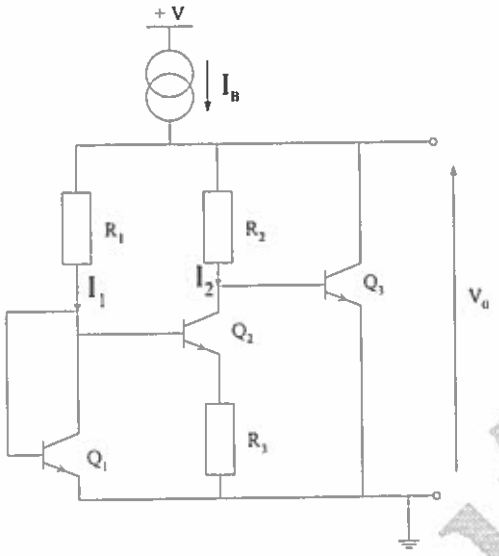
Assume V_{BE}=0.7

$$TC_F = \left(\frac{1}{0.7}\right)(-2.5mV) - \left(1500x10^{-6}\right)$$

$$= -5100\,ppm/^{o}C$$

[3 points]
[Total 6 points]

c.) Bandgap voltage reference circuit which has almost zero temperature coefficient, used mainly as stable voltage reference in ICs.



[4 points for figure]

For BG reference,

$$-V_{BE1} = V_{BE2} + I_2R_3 \quad ; \beta \rangle\rangle 1$$

$$V_{BE1} - V_{BE2} = V_T \ln\left(I_1/I_2\right)$$

$$V_o = V_{BE3} + \left(R_2/R_3\right)V_T \ln\left(I_1/I_2\right) \quad ; V_{BE3} = V_T \ln\left[I_3/I_s\right] \; assume\; room\; temp$$

$$For \;\; \partial V_o/\partial T = 0 \Rightarrow \partial V_{BE3}/\partial T = \frac{V_T}{T}\frac{R_2}{R_3}\ln\left[\frac{I_1}{I_2}\right]$$

$$Since \;\; \partial V_{BE}/\partial T = -2.5\,mV/^{o}C \;\; and \;\; \frac{V_T}{T} = \frac{1.38\times10^{-23}}{1.6\times10^{-19}}$$

$$\Rightarrow \left(\frac{R_2}{R_3}\right)\ln\left(\frac{I_1}{I_2}\right) = 29 \Rightarrow V_o = 1.283$$

[4 points for derivation]
[Total 8 points]

<u>Answers to question 2</u>

a) Figure 2.1:

Solutions

Advantage: Large voltage headroom/good for low voltage operation as has needs only one ΔV on output to be in saturation.
Disadvantage: Errors on output current due to ΔVDS will be significant.

Figure 2.1:
Advantage: Improved output impedance due to cascoding and therefore less error due to early effect, changes in ΔVDS
Disadvantage: Two output transistors reduce output voltage headroom because circuit needs two ΔV to keep output transistors in saturation.

[1 point for each, total 4 points]

b) From notes:

Improved Biasing for Maximum Output Swing (2)

Assuming Equal lengths and $I_O = I_{IN}$ Remember: β = KW/2L

$$I_D = \beta(V_{GS} - V_T)^2$$

$$\beta_1 = \beta_2 = \beta_4 = \beta_5 = \beta_6 = \beta$$

$$\beta_3 = \left[\frac{\beta}{4}\right]$$

Since equal I_D, then for Q_3, $V_{GS3} = 2(V_{GS} - V_T)$

EXAMPLE

Let $(V_{GS} - V_T) = \Delta V$ and prove to yourselves that

$$V_{OUT(min)} = 2\Delta V$$

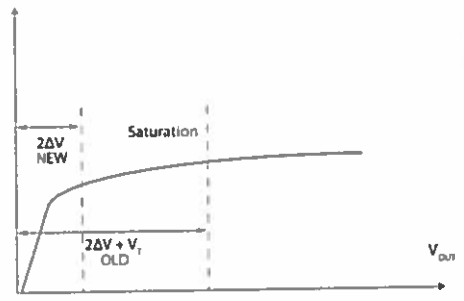
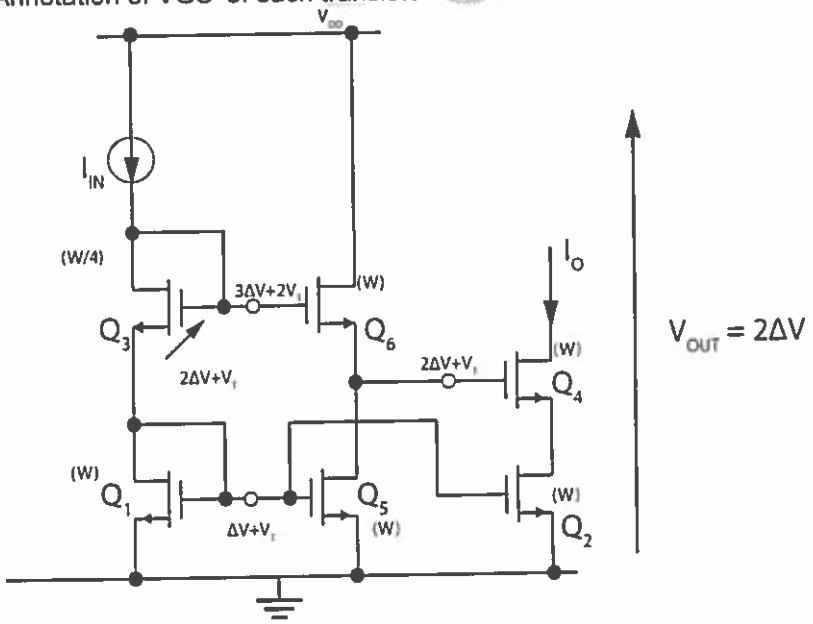


Fig. 32

HINT: remember that $V_{DS} = V_{GS} - V_T$ or $V_D = V_G - V_T$

Annotation of VGS' of each transistor:



[Total 6 points]

c) Current feedback amplifier

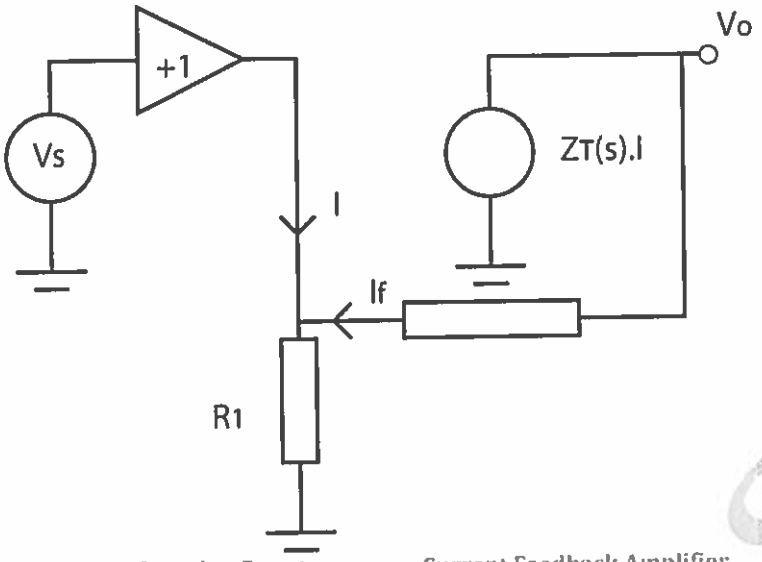


Figure 1. Question 5 part c answer; Current Feedback Amplifier.

$$Z_T(s) = Z_{T0} \Big/ \left(1 + jf/f_p\right)$$

Where fp is the dominant pole

From model

$$I_f = \left(V_o - V_s\right) \Big/ R_f$$

$$I_1 = V_s \Big/ R_1$$

$$V_o = Z_T I = Z_T\left(I_1 - I_f\right)$$

$$V_o \Big/ V_s = \left(1 + R_f \Big/ R_1\right)\left[\frac{Z_T(s)}{R_f + Z_T(s)}\right] = \left(1 + R_f \Big/ R_1\right)\left[\frac{Z_{T0}}{R_f + Z_{T0}}\right]\left(\frac{1}{1 + j f \Big/ f_p\left(\frac{Z_{T0} + R_f}{R_f}\right)}\right)$$

Assuming ZT0>>Rf

$$\left(V_o \Big/ V_s\right)_{j\omega} = \left(1 + \frac{R_f}{R_1}\right)\left(\frac{1}{1 + jf \Big/ GB \Big/ R_f}\right) \quad ; GB = f_p Z_{T0}$$

$$f_{p-closed} = GB \Big/ R_f$$

$$A_{closed} = \left(1 + \frac{R_f}{R_1}\right)$$

Therefore the bandwidth is defined by Rf and gain by R1

- Explaining why gain is independent of bandwidth in current feedback amplifier [10]

Solutions

Answers to question 3:

3. a.

$$Av1 = -gm2/(go2+go4)$$

$$(go2+go4) = I_{D2}(\lambda_N + \lambda_P) = 5\times10^{-6}\times0.05 = 2.5\times10^{-7}\Omega^{-1}$$

$$gm2 = 2\sqrt{\beta_2 I_D}$$

$$\beta_2 = \frac{K_N}{2}\left(\frac{W}{L}\right)_2 = 7.5\times10^{-5} A/V$$

$$gm2 = 3.87\times10^{-5} S$$

$$Av1 = -154.9$$

$$Av2 = -gm6/(go7+go6)$$

$$(go7+go6) = I_{D6}(\lambda_N + \lambda_P) = 20\times10^{-6}\times0.05 = 10\times10^{-7}\Omega^{-1}$$

$$gm6 = 2\sqrt{\beta_6 I_{D6}}$$

$$\beta_6 = \frac{K_N}{2}\left(\frac{W}{L}\right)_6 = 1.6\times10^{-4} A/V$$

$$gm6 = 1.13\times10^{-4} S$$

$$Av2 = -113$$

$$A_{total} = Av1\times Av2 = 17503$$

$$G.Bp = gm2/2\pi C_c = 12.32 MHz$$

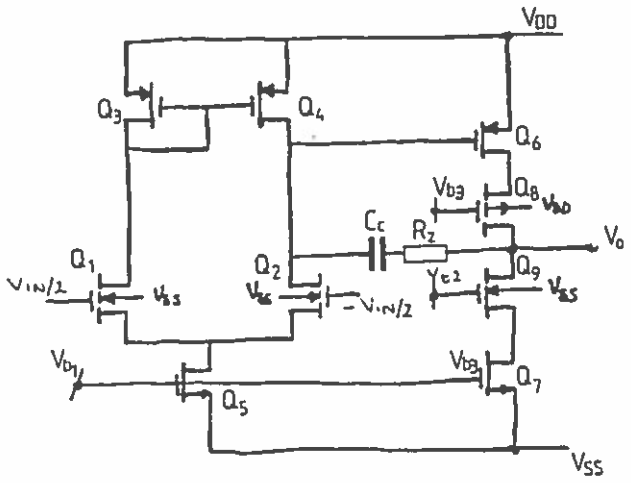
[8]

In a 2-stage opamp the load contributes to the 2nd pole hence reducing load increases stability.
With a single-stage, the load forms the dominant pole hence reducing the load increases bandwidth.

[2]

c.

2. CASCODED OUTPUT



[6]

Cascoding output stage increases the output impedance by:
Go=[go6(go8/gm8) +go7(go9/gm9)]
Since Gain of output stage=gm6/[Go]=Av2

The overall gain therefore increases.

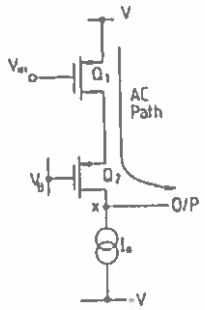
[4]

Solutions

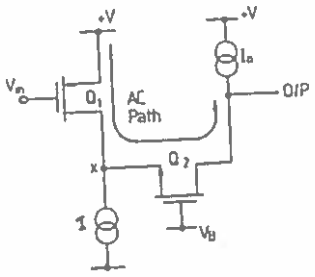
Answers to question 4:

4.a

Concept



Conventional 'stacking' cascode

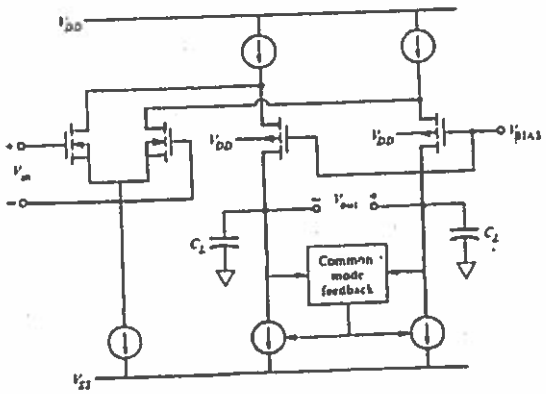


Folded cascode

In the folded cascode we are unstacking the conventional cascode and spreading it out. The AC current path is folded and this allow a reduction in power supply. The conventional cascode requires a 2-stage architecture and since the impedance at (x) is high requires internal compensation. The folded cascode can be used as a single stage architecture, node x is low impedance so the only high impedance node will be at the output.
Gain of the folded cascode Av=gm1/Go.

[4]

b.



[6]

c)

Circuit is a differential continuous time integrator with a balanced double differential linear active transresistor.

$$I_1 = 2\beta[(V_g - V_x - V_T)(V_{in}/2 - V_x) - 1/2(V_{in}/2 - V_x)^2]$$

$$I_2 = 2\beta[(V_g - V_x - V_T)(-V_{in}/2 - V_x) - 1/2(-V_{in}/2 - V_x)^2]$$

$$R = 2V_{in}/(I_1 - I_2)$$

$$R = \frac{1}{\beta(V_g - V_T)}$$

[5]

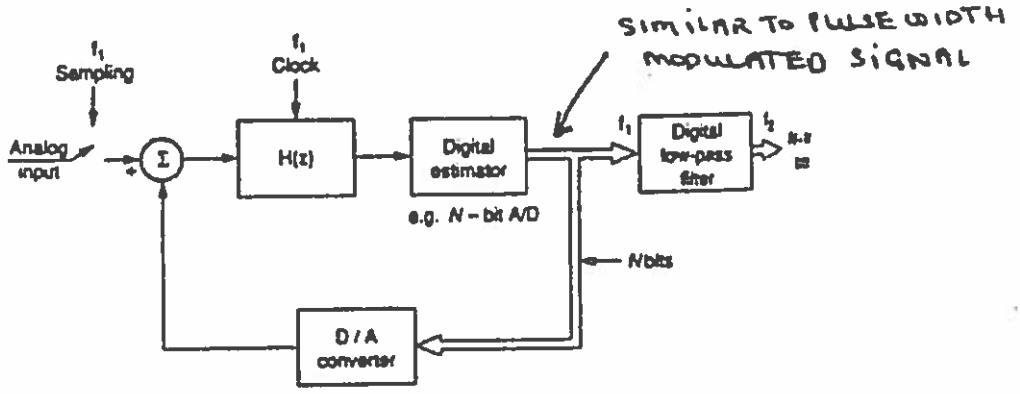
Time constant τ=RC=10ms
R=1/20ux5(1-0.5)=20 KΩ
C=τ/R=10m/20K=5x10^{-7} f

[5]

Answers to question 5:

5.a) Explain sigma delta operation (bookwork).

[5]



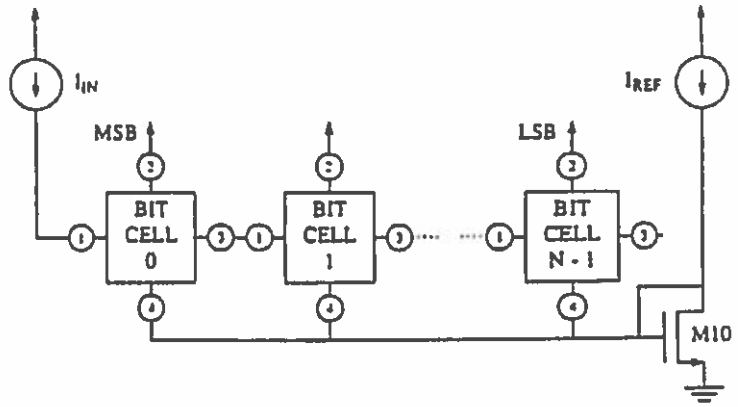
Typical oversampled Σ-Δ (modulator) converter architecture .

[5]

b)(i) Explain how the conversion works (bookwork)

[3]

Draw a cascade of blocks whereby N=8



[4]

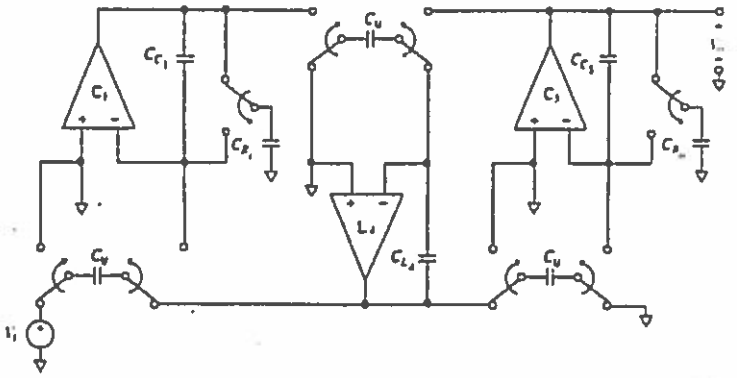
c) KT/C noise limits the resolution of a sampled data converter.

$$DR = 2^N = V_{ref}/Noise = V_{ref}/\sqrt{(kT.10Rf_c)}$$

[3]

Answers to question 6:

6.a. 3rd-order Chebyshev low pass switched-capacitor ladder filter.



[5]

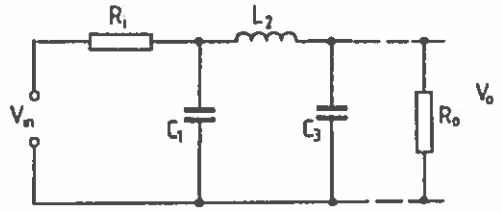
b. General transformation rules for ladder prototypes:

Inductor: $\frac{f_c L_2}{R_s} = \frac{C_{L2}}{C_u}$

Capacitor: $\frac{C_{c3}}{C_u} = f_c R_s C_3$

Resistor Rs=dummy scalar.

The circuit is equivalent to an RLC prototype



[5]

c. For switched capacitor equivalent:

C_{c1}=Cc3=5.08 pF

C_{L2}=3.49 pF.

C_u=1pF

Assume scaling Rs=Ri=Ro=1Ω

Therefore

L2= CL2/fc= $3.49/100\text{x}10^3=3.49\text{x}10^{-5}$ H

Normalised 1rad/sec we multiply by $2\pi f_0$

L2=$3.49\ \text{x}10^{-5}$ x 2π x 5 KHz =1.096 H

C1=C3=Cc3/fc = $5.08/100\text{x}10^3 = 5.08\text{x}10^{-5}$ f

Normalised value C1=C3=1.596